

SonarAuth: Using Around Device Sensing to Improve Smartwatch Behavioral Biometrics

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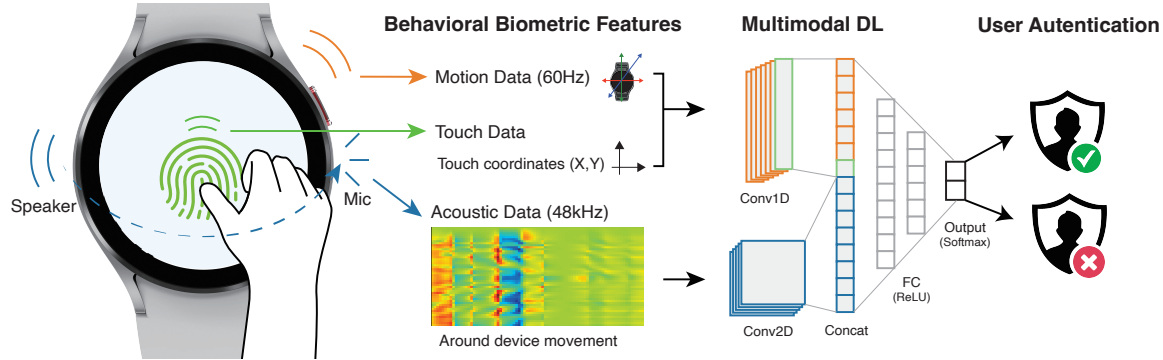


Figure 1: SonarAuth system. A user taps their smartwatch and behavioral biometric features are extracted. In addition to standard touch and motion features, we capture the motions of the hand over the watch during input using active sonar. A structured inaudible ultrasonic signal is emitted from the speaker on one side of the watch, transformed by its passage through the environment, and captured by a microphone on the other side. We process this signal to identify distinctive hand poses and movements that can enhance the performance of smartwatch behavioral biometrics.

ABSTRACT

Smartwatches are used by millions of people for applications in health, finance, and communication. As the computational power and range of applications supported by these devices expand, it is becoming more and more important to secure access to them. While various user authentication technologies have been extensively explored in smartphone use scenarios (e.g., FaceID, fingerprint, PIN, or pattern) the applicability of these approaches to smartwatches is typically limited due to the small watch form factor. To improve authentication on smartwatches, we propose SonarAuth, a novel user authentication system for unmodified commercial smartwatches using behavioral biometrics derived from motion, touch, and around-device motions. To capture in-air hand motions, we adapted an existing sonar system for smartwatches. We collected data from 24 participants from single touch to the watch screen with the thumb, index, and middle fingers. Using a multimodal deep learning classifier, we achieved a promising mean Equal Error Rate(EER) of 6.41% for user authentication based on a single thumb tap. We note that our system is usable and has good potential to be combined with other authentication modalities.

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CCS CONCEPTS

• Security and privacy → Biometrics.

KEYWORDS

behavioral biometrics, sonar, smartwatch

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1 INTRODUCTION

Smartwatches are reliable, capable, and powerful wearable computers that are used by millions. Their close proximity to the body and readily availability make them ideal for diverse applications: capturing, storing, and analyzing health and fitness data [1, 23], providing messaging and notification services [16], storing personal credentials [21] and making financial transactions [11, 18]. Such applications inevitably require access to highly sensitive personal information and services. As such, it is important to secure access to smartwatches by, for example, implementing authentication techniques such as lock screens. Indeed, this is becoming common practice. PIN and/or pattern [19], two device lock schemes popular on smartphones, have long been built into the smartwatch

platforms. The use of authentication technologies is also becoming mandatory on smartwatches for the purpose of financial services.

However, while the principles are sound, the practice is more challenging. Users may be reluctant to enable lock screens due to the burden involved in authenticating [13, 22]. Moreover, entering complex information precisely on the small screen of a smartwatch can be slow or error prone [22] due to issues such as the "fat-finger" problem [12, 24]—users' own fingers obscure and occlude the touch screen, interfering with input tasks. Consequently, if users do opt to enable lock screens, they often select credentials that are deliberately easy to enter and remember [2, 27]. Unfortunately, such commonly selected passcodes offer only limited security benefits: they are notoriously easy to guess [17]. In addition, if authentication occurs in public spaces, then knowledge-based approaches, such as PIN and pattern, are readily observable by both attackers who may be present in person [28] and by those who have digitally surveilled an area [26].

Biometrics offers potential solutions to these problems: a biometrically secured device can only be unlocked if it recognizes certain characteristics or features of its owner. Such techniques, in the form of fingerprint readers [20] and facial recognition systems [3, 25] have become popular, even dominant [2] solutions for smartphone authentication. Despite their success, these established technologies have rarely been integrated into smartwatches due to the physical (e.g., device surface area occupied), computational or battery resources required. It remains difficult to integrate and power dedicated biometric sensors in a watch form factor device. Consequently, there is growing research interest in using behavioral biometric authentication techniques for smartwatches. These techniques typically re-purpose existing device sensors, such as the touch screen [14] or motion sensors [9], to capture detailed behavioral profiles of users during either sustained or specific interactions. These profiles capture aspects of how users touch, hold, and move their devices that are sufficiently stable and distinct to support effective and accurate authentication. Evidence to support these assertions is mixed: while authentication accuracies can be high (e.g., EER: 1.37% [15]), they are not universally so (e.g., EER: 12.7% [5]). In addition, it may be relatively easy to imitate observed behaviors [8], suggesting these techniques may provide only limited resistance to observation.

To address these limitations, we propose augmenting smartwatch touch biometrics with additional data describing user behavior. Specifically, we focus on including data reflecting unencumbered around-device interaction [6]: the finger and hand poses and motions that occur above and around a watch during input tasks. We achieve this on a commercial smartwatch by adapting a recently proposed sonar system [10]—we emit structured inaudible ultrasonic sounds from the watch's speaker, mounted on one side of its body, and capture responses on its microphone, mounted on the opposite side. These signals are altered by their passage through the air, over the watch, and past the user's fingers and hands. With this system, we conducted a study capturing data from 24 participants performing simple single-tapping tasks: single watch taps with three different fingers (thumb, index, and middle). By exploring this diversity of tasks, we sought to explore what kinds of input generate the most stable and distinguishable sonar responses. The results indicate a single thumb tap shows the best performance to

authenticate a user with a mean performance of 6.41% average EER, followed by 11.15% for middle finger tap and 13.78% for index finger tap. Considering that SonarAuth relies on just a single finger tap, we believe it is usable on smartwatches and that its performance could be enhanced in input scenarios involving a sequence of taps.

The contributions of this work are proposing, implementing and evaluating the first around-device behavioral biometric system for a smartwatch. In addition, our system could be implemented on other platforms equipped with IMU sensor, a speaker, and a microphone to capture behavioral biometric features.

2 SYSTEM

We developed SonarAuth as a wearOS application on an unmodified Samsung Galaxy Watch 4 smartwatch. To use SonarAuth, the user simply taps the smartwatch with a finger, and we segment captured sensor data in a temporal window around the touch. We then use multimodal deep learning to process this data and authenticate the user. Figure 1 illustrates the overall SonarAuth system.

2.1 Data Collection

We captured touch coordinates, device movements (accelerometer, gyroscope, magnetometer, gravity, linear acceleration, and rotation vector), and in-air hand motions around the touch. We logged motion and touch data from the device's screen and IMU at 60 Hz and used its speaker (mounted on the left) and microphone (on the right) to simultaneously play and record 48kHz 16-bit PCM audio sonar signals that represent objects and movements around the device. The sonar system is based on a previously deployed implementation [10]. It uses a looping up-sampled Zadoff-Chu (ZC) sequence (original length = 127, $u = 63$, up-sampled to 1024 data points) modulated into a 6kHz band over a 20.25kHz carrier wave. The signal is inaudible and provides excellent auto-correlation features to support the accurate detection of multiple proximate sonar reflections. In addition, each individual ZC sequence spans just 21.3ms, thus ensuring the system remains sensitive to rapid around-device movement.

2.2 Preprocessing

To analyze the captured audio signals, we demodulate the signal and estimate its impulse response by conducting auto-correlation against the original up-sampled ZC sequences. By concatenating the results of auto-correlation, we can generate an image illustrating the changing sonar response during the authentication tap. To avoid including extraneous data, we generate the image using ZC sequences captured immediately around the screen touch. Specifically, we capture 50 ZC sequences of data (25 before and after the finger touches the surface), representing a 1.065-second period (50 by 21.3ms) around the finger-down event. In addition, we limit the distances at which we track sonar signals to spatial regions immediately around the watch by considering the auto-correlation data only around its highest peak (representing the direct signal path between the speaker and microphone). By considering 100 samples (50 before and after the peak points), we capture data from approximately 18cm around the watch ($343(\text{m/s}) * 0.021(\text{ms}) * 50(\text{samples}) / 2(\text{reflections})$). We then remove static components of the image by subtracting the first signal from all subsequent signals. Finally, we

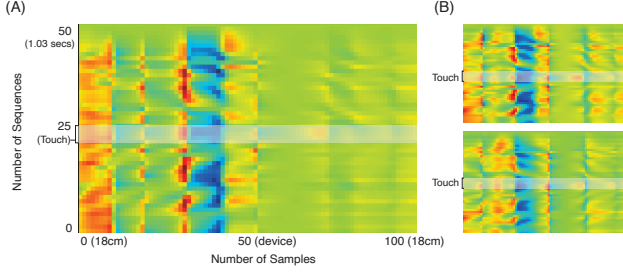


Figure 2: Examples of generated sonar image, showing the impulse response estimate from a smartwatch touch (A), and two examples of augmented data derived from the same signal using the phase rotation technique (B).

conduct data augmentation to increase our data set size. Specifically, we rely on the fact that the asynchronous speaker and microphone generate an unpredictable phase offset [4] and generated eight different sonar images from data for each tap by rotating the phase of the ZC sequences. Figure 2 illustrates an example of generated sonar image and two samples of augmented data derived from the same signal.

To analyze the motion data and touch coordinates, we first merged the data from different sensors based on the closest timestamps. In the periods before and after touch, we padded the touch coordinate data with zeros. We also windowed the data for one second around the touch-down by retaining 60 samples (30 before and after the touch-down). Finally, all motion data was normalized.

To sum up, we generated 8 augmented sonar images (50: the number of sequences, 100: the number of samples) with motion and touch biometric features (60: the number of data points, 20: the number of features) for each tapping trial.

2.3 Multimodal Classifier

We developed a multimodal classifier to combine motion, touch, and sonar images. For motion and touch data, we built a simple 1D CNN classifier consisting of two 1D convolutional layers (the number of filters: 128, 256 each, kernel size: 10, stride: 2, ReLU activation) and each is followed by max-pooling layers (pool size: 2) and a dropout layer (p: 0.5). For sonar images, we built a 2D CNN classifier consisting of four 2D convolutional layers (the number of filters: 32, 64, 128, 256 each, kernel size: (3,3), stride: (1,1), ReLU activation) with max-pooling layers (pool size: (2,2), stride: (2,2)). After flattening the output from the last convolution block from both uni-modal classifiers, we concatenate them and feed them into three fully connected dense layers with, respectively, 1024, 512, and 128 units. To validate our multimodal classifier, we also evaluate the efficacy of a single 2D CNN classifier for sonar images (directly connected to a series of fully connected dense layers after flattening). As we evaluate our system on three fingers for every user, we constructed a total of 72 (3 fingers * 24 users) authentication models. In every model, except for a genuine user, all other 23 participants are labeled as attackers. We then randomly split 23 attackers into 17 for training and 6 for testing the model. Among 8 blocks of each user’s data from the training group, the first 6

blocks are used for training, and the last 2 blocks are used for validation. We balanced the number of data between genuine users and attackers by randomly selecting a subset of data from each attacker during training.

3 STUDY

We conducted a study to collect behavioral biometric data to explore the feasibility and performance of SonarAuth. In addition to logging sensor watch data, we recorded videos of the tapping behaviors of each participant. The study was approved by the local IRB.

3.1 Participants

We recruited 24 participants (12 males, mean age of 22.13 (SD: 3.57)) from the local university. They are all right-handed, and relatively unfamiliar with smartwatches (2.13/5, SD: 1.42), compared to smartphones (5/5) or computers (4.67/5, SD: 0.56). Participants’ hand span was a mean of 18.5cm (SD: 2.88cm), their hand length was a mean of 18.2cm (SD: 1.3cm), and their palm width was a mean of 7.75cm (SD: 0.9cm).

3.2 Experiment Protocol

The experiment took place in an empty classroom with participants seated on a fixed chair in front of an empty desk. Before the experiment, participants read instructions and signed informed consent. After wearing the smartwatch on their left (non-dominant) wrist, participants completed a short practice session, in which they were asked to make a total of 30 taps to the watch, practicing this action with each of their thumb, index, and middle fingers. They were instructed that the goal of this session was to arrive at a comfortable and efficient way of performing each finger tap. After completing this session, the formal study began.

The study was composed of eight blocks of 30 trials, with ten trials in each block requiring the use of one of the thumb, index, or middle fingers. Trials in each block were randomly ordered. Each trial commenced by requesting participants tap on an on-screen button (with any finger) to begin. A centrally located trial target was then presented, visually depicting which finger to use. The participant’s task was simply to tap it with the appropriate digit. The trial then ended and the subsequent trial began. We asked participants to verbally report any errors (e.g., wrong finger use) to facilitate excluding such data from subsequent analysis. Participants were able to rest at any time between trials. Between blocks, they were required to take off and re-wear the smartwatch—this helped ensure the data captured reflected the kind of minor variations in wearing pose that would occur naturally during daily device use. At the end of the study, participants completed demographics and the NASA Task Load Index (TLX), a measure designed to assess the workload experienced during the use of SonarAuth.

4 RESULT

In total, data from 5760 trials were recorded: 10 repetitions by three fingers by eight blocks by 24 participants. Participants self-reported 11 errors (0.19%) in which they used the wrong finger to tap the targets and these trials were excluded from further analysis. For each of the remaining 5749 trials, we captured and analyzed

Table 1: Mean of individual user authentication performances for each finger tap in both multimodal and sonar image classifiers.

Classifier	Finger	EER (%)	FAR (%)	FRR (%)
		Mean (SD)	Mean (SD)	Mean (SD)
Multimodal	Thumb	6.41 (5.1)	8.02 (8.4)	4.71(5.3)
	Index	13.78(7.6)	13.94(13.1)	16.55(12.3)
	Middle	11.15(6.4)	9.57(6.4)	14.27(11.6)
Sonar Image	Thumb	14.6(9.03)	12.4(8)	16.1(11.2)
	Index	19.26(11.2)	17.21(9.3)	21.21(15.8)
	Middle	15.77(8.8)	13.6(7.7)	16.82(11.9)

touch coordinates and motion data at 60Hz and the audio signal containing the reflected structured inaudible ultrasonic signal.

4.1 Classification Performance

To evaluate our classifiers, we measure Equal Error Rate (EER), False Acceptance Rate (FAR), and False Rejection Rate (FRR). Table 1 shows overall performance in both multimodal and sonar image classifiers. The multimodal classifier shows more robust performance for the thumb tap (Mean EER: 6.41 (SD 5.1), FAR: 8.02, FRR: 4.71), followed by the middle finger tap (Mean EER: 11.15 (SD 6.4), FAR: 9.57, FRR: 14.27) and worst performance with the index tap (Mean EER: 13.78 (SD 7.6), FAR: 13.94, FRR: 16.55). The sonar image classifier shows a similar trend. Peak performance is achieved with the thumb tap (Mean EER: 14.6 (SD 9.03), FAR: 12.4, FRR: 16.1) then the middle finger tap (Mean EER: 15.77 (SD 8.8), FAR: 13.6, FRR: 16.82) and finally the index tap (Mean EER: 19.26 (SD 11.2), FAR: 17.21, FRR: 21.21). Overall, the multimodal classifier shows marked improvements in performance over those achieved with the sonar image classifier.

4.2 Usability

After the study, participants reported usability in terms of preference, comfort, and speed regarding the three finger-tap; thumb, index, and middle tap. By and large, participants marked index tap as the first in all metrics (preference: 87.5%, comfort: 79.17%, speed: 79.17%) followed by thumb tap (preference: 12.5%, comfort: 16.7%, speed: 12.5%) and middle tap (preference: 0%, comfort: 4.2%, speed: 8.3%). Although the thumb had its adherents, we also note that it was also most frequently rated as the least preferred option (preference: 66.7%, comfort: 62.5%, speed: 66.7%), suggesting participants were split in their subjective opinions of this input action. Table 2 shows the full results from this survey. Participants also completed the NASA-TLX to report their mental workload. The results were as follows: mental demand (5.17/20, SD:5.04), physical demand (4.21/20, SD:4.24), temporal demand (9.67/20, SD:6.11), performance achieved (4.5/20, SD:6.14), effort expended (5.96/20, SD:5.3), and frustration experienced (2.58/20, SD:3.63).

5 DISCUSSION AND CONCLUSION

This paper proposes a user authentication system based on behavioral biometrics derived from motion, touch, and sonar-sensed around-device movement on an unmodified smartwatch. Based on

Table 2: Usability in terms of ratings of performance, comfort, and speed across thumb, index, and middle finger taps

	preference(%)			comfort(%)			speed(%)		
	Thumb	Index	Middle	Thumb	Index	Middle	Thumb	Index	Middle
1st	12.5	87.5	0	16.7	79.17	4.2	12.5	79.17	8.3
2nd	20.8	4.17	75	20.8	16.67	62.5	20.8	16.67	62.5
3rd	66.7	8.33	25	62.5	4.17	33.3	66.7	4.17	29.2

data from a single tap on a smartwatch screen, we achieved a mean performance of 6.41% EER for thumb taps, followed by 11.15% EER for middle finger taps and 13.78% EER for index finger taps (13.78% average EER) using a multimodal classifier. A classifier using only sonar data also shows a similar trend but achieves approximately 6% lower performances overall. This suggests that a multimodal approach improves behavioral biometric features on smartwatches for user authentication.

We can make a number of other interesting observations. Perhaps most importantly, we can shed light on how the thumb tap outperforms the index and middle fingers by examining and analyzing participants' tapping behaviors, as recorded in video of the study. There are several factors that may contribute to the increased performance of the thumb. For instance, as the thumb is shorter than the other fingers, participants' hands must be located closer to the watch during thumb screen taps, which may provide more information related to the around-device movement. In addition, we note that participants showed relatively similar tapping hand poses for the index and middle finger taps (based on extending the tapping digit alone) compared to the thumb tap (which showed various articulations of the other fingers) in recorded videos. This may have made the hand poses used during thumb taps more distinctive. Participants also reported the thumb tap is the least preferable, a result that may be due to the requirement for larger gross movements of the hand. However, more positively, we note that NASA-TLX result shows that single tapping authentication, in general, places very low burdens on users in terms of mental and physical demand [7].

This work is impacted by numerous limitations. Most seriously, the peak result, 6.41% average EER for thumb tap, suggests our system is not sufficient to support effective user authentication. However, as a series of taps could be concatenated into a sequence, we note it may be possible to increase performance in future work. In addition, this paper validates the classifiers we train without separate multi-recall sessions where diverse variations in tapping behavior may occur—more prolonged empirical work is required to establish if the performance we report for SonarAuth holds in a real-world setting. Despite these limitations, our data suggest that utilizing around-device movement sensing to improve smartwatch behavioral biometrics for mobile authentication has strong potential. We achieve a promising level of performance from almost the simplest possible input action—just a tap to the screen. Examining more complex input behaviors is likely to lead to substantially improved performance and is a good topic for further study.

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